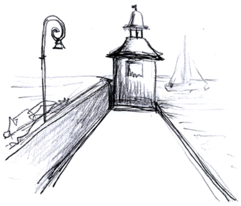

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Overview of Data Communication in the AP1000 I&C System

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ABSTRACT

A high-level overview of the Instrumentation and Control (I&C) architecture that is part of AP1000™[1] Certified Design [1] is presented along with a description of its constituent systems. The major non-safety systems are integrated using a plant-wide real-time data distribution network. Within each division of the safety system, the internal functions utilize safety datalinks and are integrated using an intradivisional safety network.

The Certified Design encompasses data flow both from the safety system to the non-safety system, and from the non-safety system to the safety system. The implementation and application of these data flow paths are described.

1 INTRODUCTION

The Certified Design [1] and the proposed amendment [2] include an integrated I&C system that will be deployed in the new construction of nuclear power plants. This paper presents the key systems of the I&C architecture and describes the functionality of each. To perform the required functions, information is communicated between the systems. The methods that are used to communicate this information are also presented.

2 AP1000 I&C ARCHITECTURE

A high-level overview of the I&C architecture is shown in Figure 1. The architecture is divided into the following systems:

- Protection and safety monitoring system (PMS) – The safety-grade protection and safety monitoring system detects off-nominal conditions and actuates appropriate safety functions necessary to achieve and maintain safety shutdown. The major subsystems within the PMS are the nuclear instrumentation system (NIS), the reactor protection system (RPS), and the qualified data processing system (QDPS). The RPS includes the reactor trip system (RTS), the engineered safety features actuation system (ESFAS), and the component logic system. The PMS is implemented using the Westinghouse Common Qualified (Common Q) Platform described in [3], and accepted by the United States Nuclear Regulatory Commission (NRC) in [4, 5, and 6].

[1] AP1000™ is a trademark of Westinghouse Electric Company LLC.

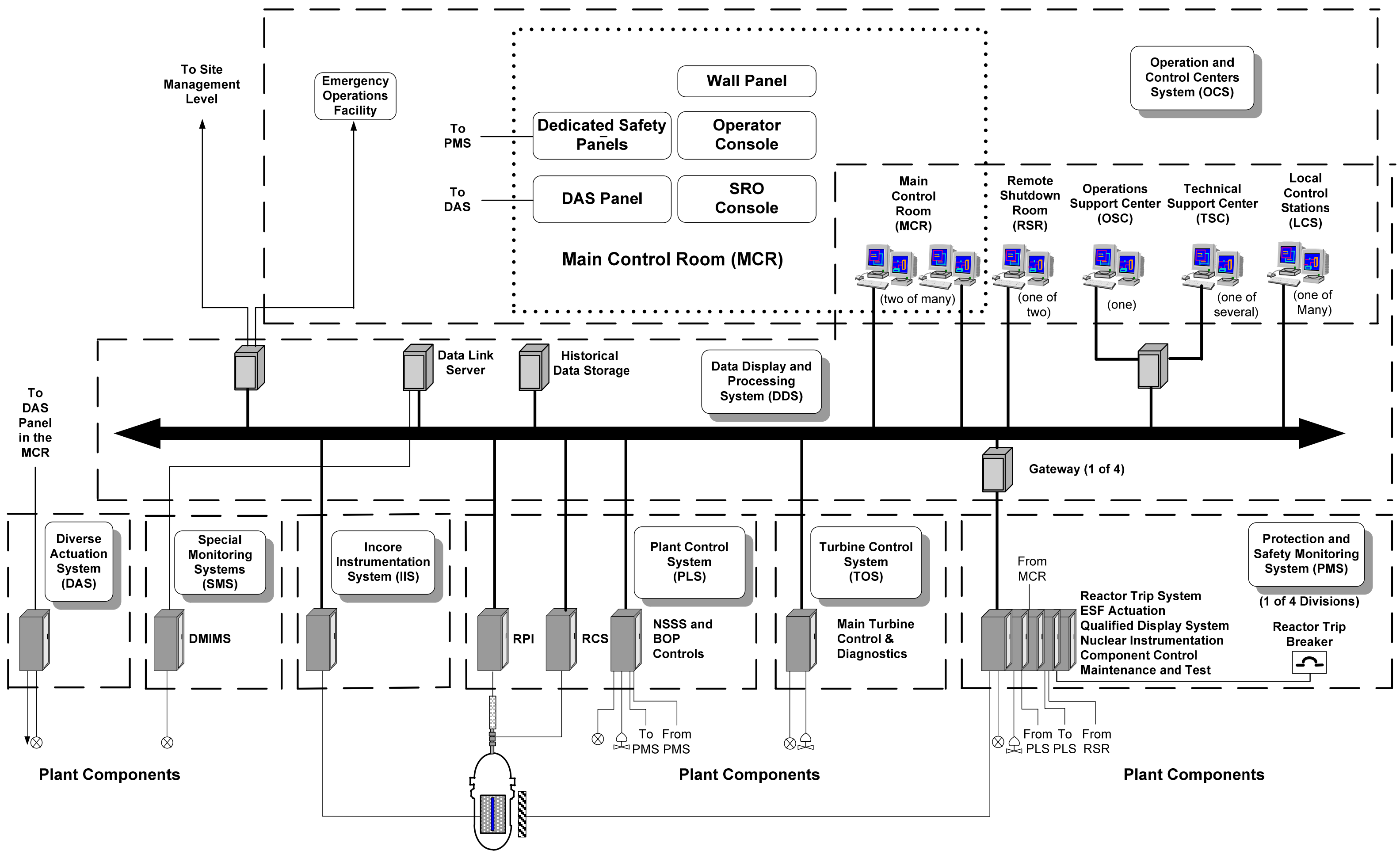

Figure 1 High-Level Overview of the AP1000 I&C Architecture

- Plant control system (PLS) – The plant control system provides the functions required for normal operation from cold shutdown through full power.

- Data display and processing system (DDS) – The data display and processing system provides data that generate alarms and displays for both normal and emergency plant operations to the equipment used for processing. It includes the displays, real-time data network, alarm system, nuclear application programs, logging function, and archiving function.

- Main turbine control and diagnostic system (TOS) – The main turbine control and diagnostic system implements the turbine protection and control functions.

- Special monitoring system (SMS) – The special monitoring system consists of standalone diagnostic systems, such as the metal impact monitoring system, that preprocess data from specialized sensors.

- Diverse actuation system (DAS) – The diverse actuation system is a non-safety system that provides an alternate means of initiating reactor trip, actuating selected engineered safety features, and monitoring plant information. This system addresses the unlikely coincidence of a postulated transient and a postulated common-mode failure (CMF) in the PMS. Although the DAS is a non-safety system, it has special design process requirements due to its mission.

- In-core instrumentation system (IIS) – The primary (non-safety) function of the in-core instrumentation system is to provide data for a three-dimensional flux map of the reactor core. Its secondary (safety) function is to provide the PMS with thermocouple signals for the post-accident inadequate core cooling monitor. The tertiary (non-safety) function of the IIS is to provide the DAS with a separate set of thermocouple signals. The safety and non-safety functions share only the mechanical instrumentation assemblies that are placed within the reactor. Separate signal processing electronics are used.

- Operation and control centers system (OCS) – The operation and control centers system includes the main control room (MCR), technical support center (TSC), operations support center (OSC), remote shutdown room (RSR), emergency operations facility (EOF), and local control stations (LCSs). From an I&C point of view, the OCS represents the integration of human-system interface (HSI) resources from the PMS, PLS, TOS, DDS, and DAS. Thus, it has portions that are safety-grade and portions that are non-safety-grade.

The major non-safety systems (DDS, PLS, TOS, SMS) and the non-safety portions of both the IIS and OCS are integrated using a plant-wide real-time data distribution network. Within each PMS division, the NIS, RPS, and QDPS functions and the safety portions of both the IIS and OCS are integrated using an intradivisional Advant®[2] Fieldbus 100 (AF100) network. This

[2] Advant® is a registered trademark of ABB Process Automation Corporation.

network is part of the Westinghouse Common Q Platform, as described in [3], and is referred to as the safety network.

In order to support the DDS and OCS functions, a large amount of data must be transferred from the safety portions of the I&C system to the non-safety portions of the I&C system. To achieve this data transfer, there are four unidirectional gateways; one for each of the PMS divisions. The Advant to Ovation®[3] Interface (AOI) communication gateways allow the PMS divisions to provide data to the non-safety systems. The AOI gateways contain a non-safety portion connected to the plant-wide real-time data distribution network and a safety system portion connected to the intradivisional network in each division. The design of these gateways meet Class 1E to non-Class 1E separation, isolation, and independence requirements.

3 NON-SAFETY COMMUNICATION

Non-safety communication occurs primarily within the non-safety communication network; a robust, fault-tolerant, high-speed, commercially available communications network designed for mission-critical process control applications. This network is implemented using the Emerson Ovation network. The non-safety communication network uses unaltered Ethernet protocols, high-speed Ethernet switches, and full-duplex cabling (fiber or copper unshielded twisted pair), and provides real-time data distribution and general-purpose communication. Real-time data distribution is defined as the scheduled periodic broadcast of real-time data pertaining to the plant processes. General-purpose communication is defined as the aperiodic exchange of data for other purposes, such as system operation, diagnostics, maintenance, etc.

Real-time data distribution within the non-safety system supports seamless integration within and among functional systems, including integration between the safety and non-safety systems.

The non-safety communication network seamlessly supports network standard communications protocols such as Transmission Control Protocol/Internet Protocol (TCP/IP) for general-purpose communication. Within the system, general-purpose communication based on TCP/IP is used for aperiodic data, including file-type data transferred from the historian and plant database to be presented at the HSI, plant informational data messages, and sequence-of-events messages to the plant historian for long-term historical storage. This communication occurs on the same physical media as the real-time periodic data, but is implemented in a way that fulfills the design philosophy which guarantees real-time periodic data transmission without loss, degradation, or delay, even during plant upsets.

4 SAFETY COMMUNICATION

Communication within the safety system consists primarily of safety datalink interfaces and the four intradivisional safety communication networks.

The PMS uses unidirectional high-speed links to serially communicate certain data within and across PMS divisions. The data consists primarily of the divisions' votes for each trip or actuation function. Unidirectional links were selected to facilitate the maintenance of communication independence between the divisions. These links are further described in [7].

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[3] Ovation® is a mark of Emerson Process Management.

Within each PMS division, the NIS, RPS, and QDPS functions and the safety portions of both the IIS and OCS are integrated using an intradivisional safety network. The safety network is a high-performance, deterministic communication network, intended for communication between controllers and flat panel display systems within the same division. Like the non-safety network, the safety network provides real-time data distribution and general-purpose communication. Real-time data distribution is referred to as “process data transfer,” and general-purpose communication is referred to as “message transfer.” Real-time data distribution is accomplished using cyclic process data transfer communication on the network.

General-purpose communication is accomplished using message transfer services. Message transfer is not performed cyclically like process data transfer, but only when one or more of the attached communication interfaces have data to send. Message transfer does not influence process data transfer in any way. Process data transfer remains deterministic. Within the PMS, general-purpose communication is used primarily for diagnostic purposes. Security is maintained, since the ability to remotely program the controllers and flat panel display systems over the network has been disabled in the PMS.

5 COMMUNICATION BETWEEN SAFETY AND NON-SAFETY EQUIPMENT

The Certified Design [1] and the proposed amendment [2] include the following types of data flow between the safety and non-safety systems:

- Data flow from PMS to PLS for control purposes – Because the PLS uses PMS sensor signal values and PMS calculated values as inputs to control functions, this type of data flow is necessary. (See, for example, Tier 2 Section 7.1.2.1 in [1].)

- Data flow from PMS to DDS for information system purposes – Because the DDS is responsible for the traditional plant computer functions that include the display, processing, alarming, logging, and archiving of PMS process and system data, this type of date flow is necessary. (See, for example, Tier 2, Section 7.1.1 in [1].)

- Data flow from DDS to PMS for safety system actuation purposes – This type of data flow is necessary to implement system-level actuation of the safety system from the RSR, which is entirely non-safety[4]. (See, for example, Table 2.5.4-2, Inspections, Tests, Analyses, and Acceptance Criteria [ITAAC] in [1].)

- Data flow from PLS to PMS for component control purposes – This type of data flow is necessary to implement soft control of safety system components from the PLS.[5] (See, for example, Tier 2, Section 7.1.2.8 in [1].)

[4] In conventional nuclear plants, the remote shutdown capability is generally implemented using safety-grade controls. That is not necessary for AP1000 because there is no regulatory requirement for safety-grade controls, and the architecture allows non-safety control of safety components. DI&C-ISG-04 [Reference 10] provides guidance for the use of non-safety controls for safety components.

[5] The plant has no credited manual operator actions. Therefore, manual component control of safety components is implemented in the PLS. DI&C-ISG-04 [Reference 10] provides guidance for the use of non-safety controls for safety components.

The required data flows are implemented using channelized unidirectional gateways and individual analog and digital signals as shown in Figure 2. Five cases are identified in Figure 2 and labeled Case A through Case E. The cases are discussed in more detail in the following sections.

5.1 Safety to Non-Safety Data Flow

5.1.1 Case A And Case B – Hardwired Signal Interfaces

Analog inputs required for both control and protection functions (e.g., pressurizer pressure) are processed independently with separate input circuitry. The input signals are classified as safety-related and are, therefore, isolated in the PMS cabinets before being sent to the PLS as individual hardwired analog signals. This type of interface is shown as Case A in Figure 2 and is identical to the type of interface used in existing plants.

The PMS also provides data to the PLS pertaining to analog and digital signals calculated within the PMS (e.g., overtemperature delta-temperature margin to trip). These signals are classified as safety-related and are, therefore, isolated in the PMS cabinets before being sent to the PLS as individual hardwired analog or digital signals. This type of interface is shown as Case B in Figure 2 and is identical to the type of interface in existing Westinghouse plants.

In both cases, qualified isolation devices are used. These devices provide electrical isolation between the systems as required by [8] and prevent all data flow (data, protocols, and handshaking) from the non-safety system to the safety system, thereby providing the communication isolation envisioned by [9], Annex G. They also provide functional isolation by preventing the non-safety system from adversely affecting the safety system.

5.1.2 Case C – Unidirectional Network Gateway

Various process-related signals (analog input signals, analog signals calculated within the PMS, and digital signals calculated within the PMS) are sent to the DDS for information system (plant computer) purposes. Non-process signals are also provided to the DDS for information system purposes. The non-process outputs inform the DDS of cabinet entry status, cabinet temperature, power supply voltages, subsystem diagnostic status, etc. Process-related signals are also sent from the PMS to the PLS that do not require the low transmission latency or control system segmentation provided by the dedicated signal interfaces described for Cases A and B.

The AOI gateway in each PMS division connects the division's internal network to the non-safety real-time data network, which supports the remainder of the I&C system. Each gateway has two subsystems. One is the safety subsystem, which is part of the PMS division and interfaces to the safety network. The other is the non-safety subsystem, which is part of the DDS and interfaces to the non-safety network. The two subsystems are connected by a fiber-optic link. This type of interface is shown as Case C in Figure 2.

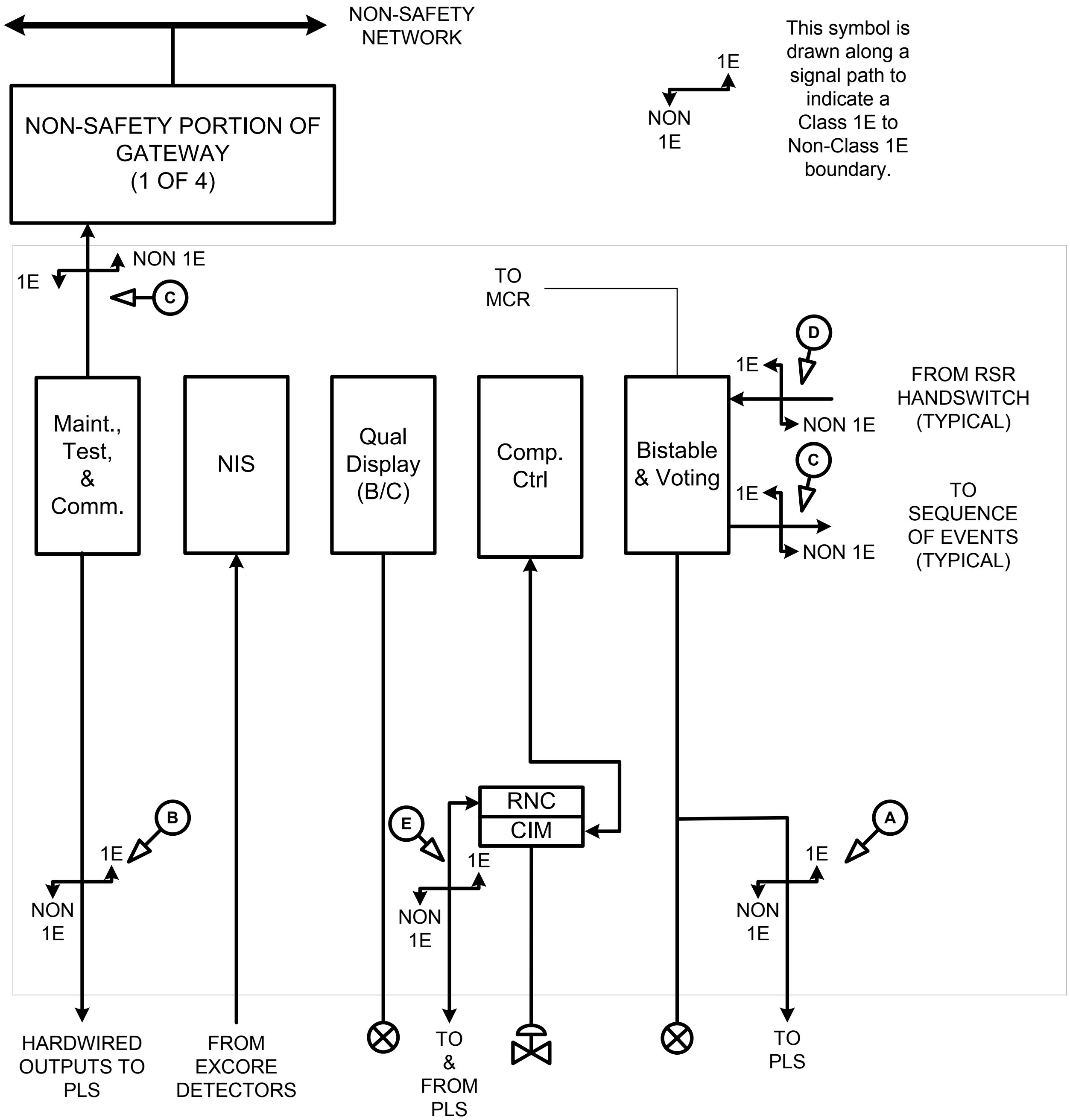

Figure 2 Data Flows Between Safety and Non-Safety Equipment

The flow of information between the two gateway subsystems is strictly from the safety subsystem to the non-safety subsystem. The unidirectional nature of the gateway is ensured by the use of a single unidirectional fiber to connect the two gateway subsystems. Within the safety system, the fiber is connected to an optical transmitter. Within the non-safety system, the fiber is connected to a fiber-optic receiver. This arrangement provides electrical isolation between the systems as required by [8] and prevents all data flow (data, protocols, and handshaking) from the non-safety system to the safety system, thereby providing the communication isolation envisioned by [9], Annex G. It also provides functional isolation by preventing the non-safety system from adversely affecting the safety function.

5.2 Non-safety to Safety Data Flow

The non-safety to safety data flows are not implemented using communication links; rather, they are implemented using discrete digital signals. These signals are used to implement non-safety manual control of system-level safety functions (actuations, manual blocks and resets, manual reactor trip) and non-safety manual component-level control of safety components.

5.2.1 Case D – Non-Safety Manual Control of System-Level Safety Functions

The non-safety manual controls of system-level safety functions (actuations, manual blocks and resets, manual reactor trip) originate from dedicated switches in the RSR. The individual hardwired digital signals are classified as non-safety-related and are, therefore, isolated in the PMS cabinets before being used. This type of interface is shown as Case D in Figure 2.

Qualified isolation devices are used to provide electrical isolation between the systems as required by [8] and prevent all but the required data flow from the non-safety system to the safety system, thereby providing the communication isolation envisioned by [9], Annex G.

Functional isolation provided by logic within the PMS prevents this data flow from inhibiting the safety function. First, the functionality associated with these controls is disabled until operation is transferred from the MCR to the RSR. Thus, these controls are nearly always disabled. This transfer is accomplished by the divisionalized Class 1E transfer switches, which are connected directly to the local coincidence logic controllers in each division. Additionally, when the controls are enabled, their functionality is limited to that defined in the PMS functional design. Specifically, the manual system-level engineered safety feature (ESF) actuations and manual reactor trip inputs can only initiate safety functions, not inhibit them. The manual system-level blocks are subject to initiation permissives and to automatic removal. The manual system-level resets only remove the system-level actuation signals; they do not cause any components to change state. An additional signal is required to cause a component to change state.

5.2.2 Case E – Non-Safety Manual Component-Level Control of Safety Components

The manual component soft controls originate in the PLS. To reduce the number of signals (cables) that must be run from the non-safety system to the safety system, the non-safety system's remote input/output (I/O) capability is used to deliver the signals to the safety system. Specifically, a remote I/O node from the non-safety system is physically located within each division of the safety system. The remote I/O node is electrically isolated from the non-safety system by the fiber-optic remote I/O bus. The node is powered by the safety system and the portions of the node not performing a safety function are qualified as associated Class 1E equipment. This type of interface is shown as Case E in Figure 2.

The remote I/O node includes one or more Class 1E component interface modules (CIMs). Internally, these modules contain the equivalent of a digital output module. The resulting digital output signals, corresponding to the demands from the non-safety system, are made available to non-processor-based priority logic also contained in the CIM. The priority logic within the CIM combines the non-safety demands with Class 1E automatic actuation signals and Class 1E manual actuation signals from the PMS. If conflicting demands are present, the safe state of the component takes priority. The CIMs also contain the equivalent of a digital input module which is used to read component status and internal CIM status. This information is made available to the non-safety system. Therefore at the point of interface to the priority logic, there are two unidirectional data flows: (1) Demands going from non-safety to safety, and (2) Status going from safety to non-safety. Each of these data flows is implemented as simple digital signals, not as a communication link.

As previously discussed, the remote I/O bus used to connect the non-safety system to the associated Class 1E remote node is fiber-optic. This arrangement provides electrical isolation between the safety system and the non-safety system as required by [8]. The remote I/O node controller (RNC) and the communication function within the CIM implement the communications, and only the resulting digital signals interface with the Class 1E priority logic in the CIM. The simple discrete signal interface within the CIM provides the communication isolation envisioned by [9], Annex G. Although the remote I/O bus uses bidirectional communication, the simple discrete signal interface between the communication function and the Class 1E priority logic ensures that the only data reaching the logic are the intended commands. The priority logic within the CIM provides functional isolation by implementing safe-state-based priority and by only implementing the functionality defined in the PMS functional design. This implementation is shown in Figure 3.

6 CONCLUSIONS

This paper describes the overall integrated I&C architecture and its constituent functional systems. It identifies the five types of data flow between the safety system and non-safety systems required by the Certified Design, and describes the implementation of each data flow. The implementations are shown to meet the requirements of [8] and [9].

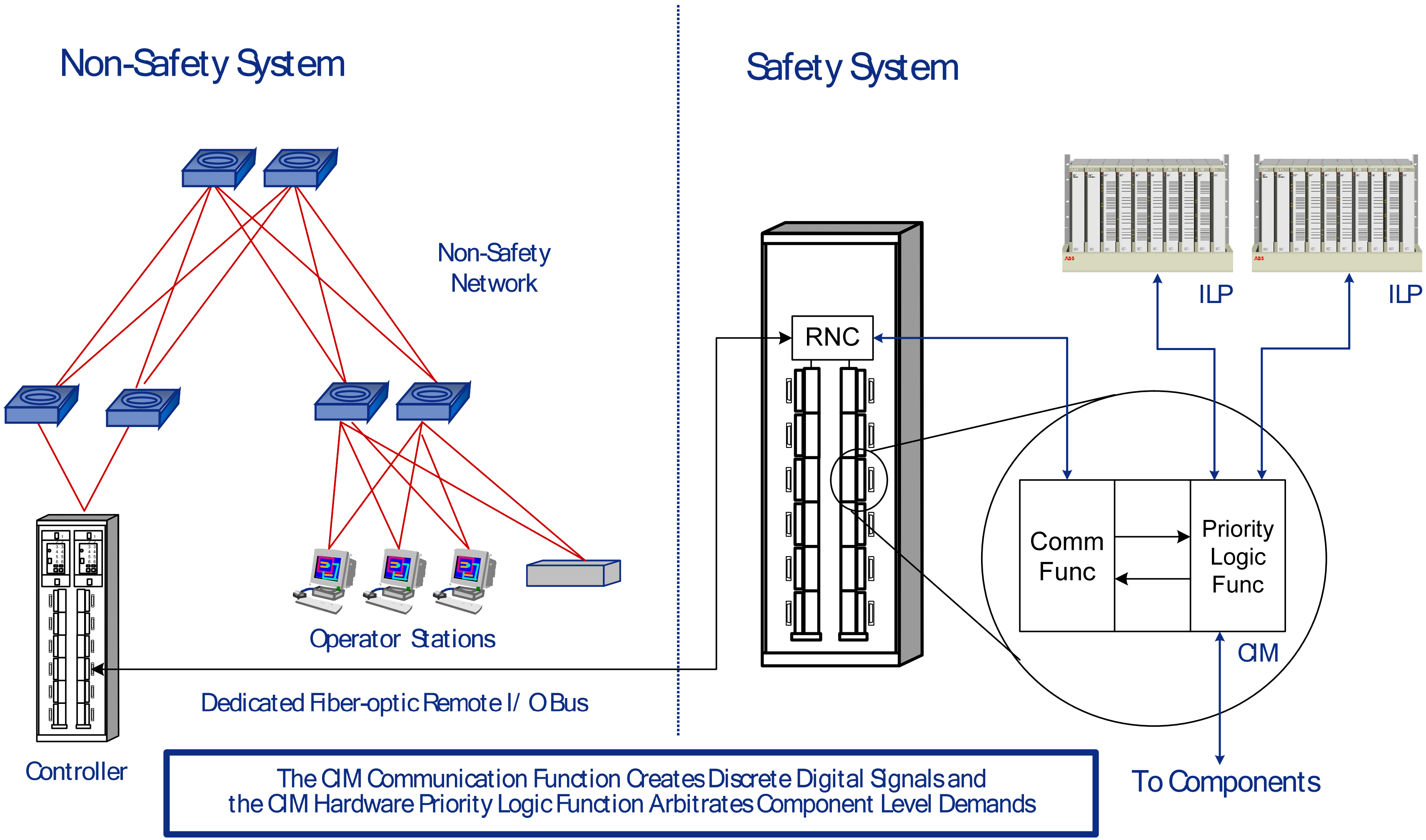

Figure 3 Implementation of Case E Data Flow

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